

KISHAN DEKA

DATA ANALYST | DATA SCIENTIST | PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

Astrophysics PhD researcher transitioning to industry data roles, with a strong analytical background and 6+ years of Python experience. Proven ability to develop elegant solutions to complex, data-intensive problems. Skilled in end-to-end pipelines, large-scale data processing, Bayesian inference, and machine learning. Collaborative communicator experienced in presenting technical findings to international audiences.

EXPERIENCE

National Centre for Nuclear Research (NCBJ) - PhD Researcher, Astrophysics

Warsaw | 4 years (present)

Bayesian inference | Monte Carlo simulations | MCMC | Noise reduction | Automated workflow

- Built end-to-end pipeline for simulation and analysis workflows spanning synthetic sky generation, signal and noise modeling, reconstruction, and validation of systematic bias.
- Developed noise-reduction and component-separation analyses, quantifying how foreground residuals propagate into downstream estimates and decisions.
- Processed large simulation outputs and applied Bayesian and MCMC methods to estimate parameters, quantify uncertainty, and test robustness under systematic effects.

National Institute of Science Education and Research - Master's Researcher

Bhubaneswar | 2 years

Image processing | Chi-square minimisation | Python package development

- Developed PSF (Point Spread Function) photometry pipeline to automate source extraction and flux measurements..
- Engineered image processing pipelines for raw observational datasets, reducing background noise contamination and optimizing signal-to-noise ratios (SNR).

SKILLS

Analytics & Statistics: exploratory analysis, hypothesis testing, A/B testing, Bayesian inference, MCMC, uncertainty quantification

Data & Pipelines: large-file processing, live data streaming, data validation, feature pipelines, end-to-end simulation, reproducible workflows

Machine Learning & Signal Processing: classification, CNNs, Image processing, noise reduction, component separation, model evaluation, bias and error analysis

Technical: Python (pandas, scikit-learn, PyTorch), C++, Fortran, SQL, Bash, Git, HPC, Streamlit, GitHub Actions

Communication: English (B2 level), international cross-team collaboration, technical presentations to broad audiences

PROJECTS

Customer retention decision analytics | *SQL analytics | PCA | Machine Learning | SHAP*

- Identified high-risk customer segments and key churn drivers using SQL analytics and predictive modeling, enabling more focused retention targeting and prioritization of revenue at risk.
- Converted churn predictions and SHAP insights into actionable retention recommendations, linking customer risk factors to targeted interventions and business decision-making.

PayWatch : Real-time transaction anomaly and fraud detection | *Stream ingestion | Hybrid ML scoring | SQL monitoring*

- Two-stage streaming fraud pipeline that combined LSTM anomaly detection autoencoders with XGBoost scoring.
- Containerized production stack with rolling SQL features, drift monitoring, asynchronous logging and automated tests.

DisasterLens : Satellite & disaster imagery classifier | *Computer vision | CNNs | Streamlit | CI/CD*

- End-to-end CNN pipeline for disaster satellite imagery; evaluated Precision, Recall, and F1-scores; reduced false negatives.
- Diagnosed confusion patterns and edge cases, automated CI/CD with GitHub Actions, and deployed a Streamlit application for real-time model inference.

EDUCATION

Ph. D. in Astrophysics | National Centre for Nuclear Research (Thesis Defense in September)

Warsaw | 2026

Masters in Physics | National Institute of Science Education and Research | GPA : 8.83/10

Bhubaneswar | 2022

Bachelors in Physics | National Institute of Science Education and Research

Bhubaneswar | 2020